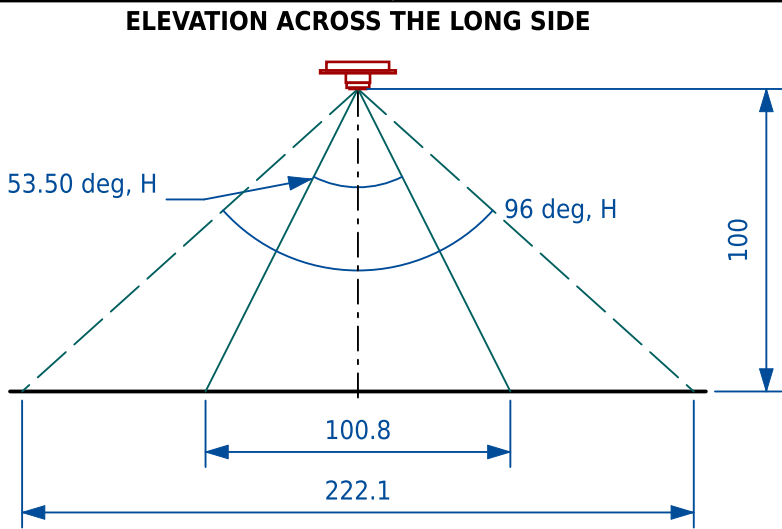
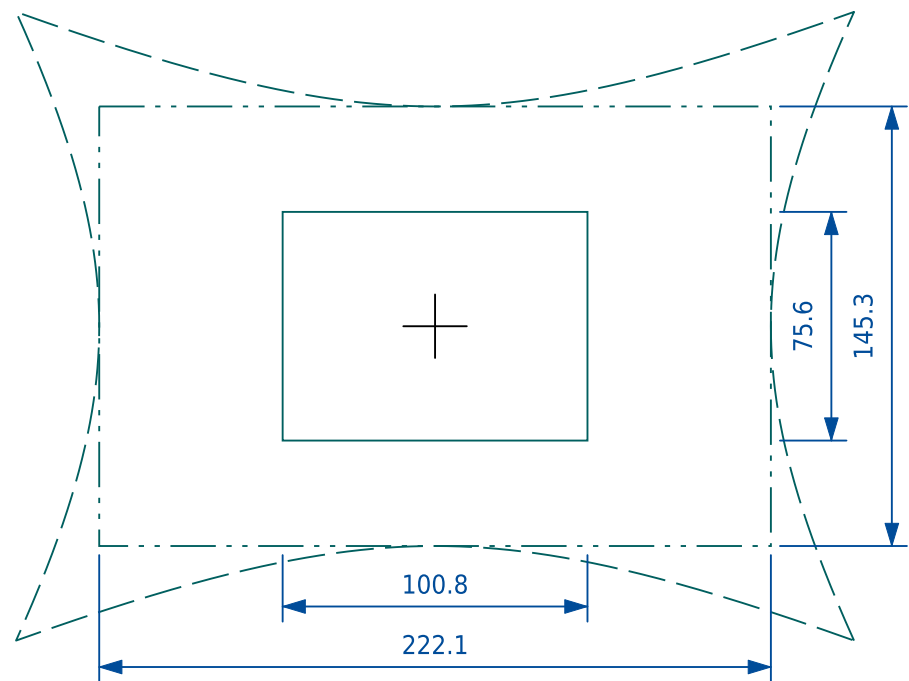




PLAN: THE PICTURE ON A BOARD 100 BELOW



NOTES (continued)

6. FIXED FOCUS, ASSUMED at 2 m, the stock lens's hyperfocal distance; the 120's is 0.70 m, so its 1 m reads copied. At 100 mm a point spreads to 30 px (1.1 mm on the board) at 65 and 13 px (0.8 mm) at 120: out of focus, and within two pixels the same set at infinity.
7. AF: no close limit is published. The others, each its own module's: the v1.3's lens unscrewed by hand "about 6 cm", 3 cm as it falls out, by forum users, Raspberry Pi publish none; the B0031 "less than an inch"; the B0176, a different motorised module, "80mm to infinity".
8. WS G, Waveshare's RPi Camera (G): 160 deg diagonal on Waveshare's page, 120 across on The Pi Hut's; its 3.15 mm gives 82.5 diagonal equidistantly, so it does not hold together and is not drawn.

NOTES

1. 65: rectilinear; its 53.50 x 41.41 come back out of 3.60 mm on the 3.6288 x 2.7216 mm active array to 0.01 deg. The 65 it is sold as is the diagonal, 64.42 on the array and 64.94 on OmniVision's image area.
2. AF: the AliExpress "AF-65 Degrees" module used here, a v1.3 clone board, not an Arducam. Its one figure is the 65, the stock lens's diagonal; its angles, f and F are the stock lens's, ASSUMED.
3. 120: the page's 120 is the DIAGONAL, so the catalogue's 120 x 90 is REJECTED: no lens sees as far across as to the corner. Its 96 x 72, for the camera without IR filter, is the diagonal x 0.8 and 0.6 like its neighbours: an equidistant split, not a measurement. YXF print D as H, H as V, V as D. The margin absorbs every row marked margin.

NOTES (continued)

4. DISTORTION: a ray A/2 off the axis meets a board Z below at Z tan(A/2) whatever the lens did to it. The dashed outline is the sensor's edge walked onto the board equidistantly: its sides bow outwards, so the rectangle the angles give, which every height is worked from, is inside the picture: its corner reaches 138.6, 103.9 against the rectangle's 111.1, 72.7. At the long side's edge a pixel covers 2.23 x the board it does on the axis.
5. DEPTH OF FIELD, DERIVED, thin lens, circle of confusion 2 px, 2.8 um: hyperfocal $f^2 / (N c) + f$, sharp from half of it to infinity focused there. Raspberry Pi's "Approx 1 m to infinity" is a 1.6 px circle.

SOURCES

1. Raspberry Pi camera documentation; OmniVision OV5647 datasheet; Arducam 5MP OV5647 documentation; Arducam B006604 product page; Arducam B0121 product page, the B0176's predecessor; Arducam B0176 on UCTRONICS, Arducam's own store; Arducam motorized focus camera; Commonlands, OV5647 lens table; YXF YXF4Y001A1 M6 lens for OV5647 modules; Waveshare RPi Camera (G); AliExpress listing of the autofocus module used here; Raspberry Pi forums, "setting lens focus", June 2013; Raspberry Pi forums, "Camera with Macro focus", November 2013; Arducam B0031, OV5647 with an M12 lens focused by hand. Addresses, pinned Internet Archive captures where one exists, in raspberry_pi_camera/optics.py; every quote checked against the cached page by raspberry_pi_camera/verify_optics.py, in ASCII here: x, um, deg and infinity for the pages' signs.

FIELD OF VIEW, deg, EVERY FIGURE

	FIGURE		H	V	DIAG	
65	Raspberry Pi	DECL	53.50	41.41	--	USED
	Arducam B0033	DECL	54	41	--	margin
	3.60 mm on the active array	DERI	53.50	41.41	64.42	agrees
AF	AliExpress, AF-65	DECL	--	--	65	
	the stock lens's pair	ASSU	53.50	41.41	--	USED
120	Arducam B006604 page	DECL	--	--	120	
	Arducam catalogue, B006604	DECL	120	90	--	REJECTED
	Arducam catalogue, B006604N	DECL	96	72	--	USED
	120 diagonal, equidistant	DERI	96	72	120	agrees
	120 diagonal, rectilinear	DERI	108.36	92.20	120	margin
	120 diagonal, equisolid	DERI	94.31	69.83	120	margin
	Commonlands CIL282, 2.2 mm	DECL	96	72	122	agrees
	Commonlands CIL282, scaled to 120	DERI	94.43	70.82	120	margin
	YXF4Y001A1, 1.79 mm	DECL, relabelled	92.40	73.90	119.90	margin

FOCUS; HYPERFOCAL AND NEAR AT 2 / 1 px

	f mm	F	FOCUS	RANGE, mm	HYPER m	NEAR m
65	3.60 DECL	F2.9 DECL	Fixed	1000 to inf	1.60 / 3.20	0.80 / 1.60
AF	3.60 ASSU	F2.9 ASSU	Motorized	not published	1.60 / 3.20	0.80 / 1.60
120	2.17 DERI	F2.4 ASSU	Fixed	1000 to inf	0.70 / 1.40	0.35 / 0.70
WS G	3.15 DECL	F2.35 DECL	Adjustable	100 to inf	--	--
v1.3 unscr.	--	--	By hand	60 to inf forum	--	--
B0031	--	--	By hand	25.4 to inf DECL	--	--
B0176	--	--	Motorized	80 to inf DECL	--	--

AF, IF FOCUSED AT Z: SHARP FROM, TO, mm

Z	NEAR	FAR	DEPTH
30	29.5	30.5	1.0
80	76.3	84.0	7.7
150	137.4	165.1	27.8

LEGEND

- The 65 deg lens: rays, and its picture on the board
- - - The 120 deg lens: rays, and its picture on the board
- - - The rectangle the 120 deg lens's angles give
- Camera Module v1.3
- Dimension, extension and leader

OV5647 LENSES AND FOCUS

Camera Module v1: stock, autofocus and 120 degree lenses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-467-gc00bb8e	mm	1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-LENS	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
OV5647 lenses	angles as declared, DERI and ASSU as flagged		